

Metadata S4

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Interannual controls of the phytoplankton assemblages in Baffin Bay

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- Description of:

Data S4: Forcing fields of silicate and nitrate concentrations from January 1 to May 15 for the reference simulation.

Simulation output for the reference simulation (EXP-0) and for the 10 sensitivity simulations discussed in more detail in Benoît-Gagné et al.: Interannual controls of the phytoplankton assemblages in Baffin Bay (in prep) (EXP-S02N02, EXP-S02N08, EXP-S02N16, EXP-S08N02, EXP-S08N08, EXP-S08N16, EXP-S14N02, EXP-S14N08, EXP-S14N16 and EXP-S14N20). The model simulations were configured in one dimension to represent the location of the ice camp of the Green Edge mission in western Baffin Bay at 67.48°N, 63.79°W in 2016. Simulations were run 10 years of 365 days. Only the last year was used for the results shown in the paper.

Each directory contains the output of one experiment. The naming convention for the directory is run_20240930_0000_EXP2_3si_yy_no3_xx_122_sims where yy is the winter silicate concentration at 100 m ($\mu\text{mol Si L}^{-1}$) and xx is the winter nitrate concentration at 100 m ($\mu\text{mol N L}^{-1}$). run_20240930_0000_EXP2_3si_standard_no3_standard_122_sims is the directory for the reference simulation (EXP-0). Each directory contains a file named car.0000000000.t001.nc.

The variables have four dimensions. Dimension T is the time. It has a size of $365 \times 10 = 3650$. The index of the dimension T is the day of the simulation. For example, the results for the tenth year are from indices $365 \times 9 = 3285$ to $365 \times 10 - 1 = 3649$ (inclusively and 0-based). Dimension Zmd000075 is the index of the depth. The correspondence between the index of Zmd000075 and the depth is given in variable RC of file grid.t001.nc:

- 1, -0,506 m
- 2, -1,533 m
- 3, -2,595 m
- 4, -3,701 m
- 5, -4,866 m
- 6, -6,107 m
- 7, -7,444 m
- 8, -8,903 m
- 9, -10,515 m
- 10, -12,315 m
- 11, -14,338 m

12, -16,623 m
13, -19,204 m
14, -22,109 m
15, -25,357 m
16, -28,951 m
17, -32,884 m
18, -37,133 m
19, -41,668 m
20, -46,455 m
21, -51,454 m
22, -56,630 m
23, -61,948 m
24, -67,381 m
25, -72,904 m
26, -78,497 m
27, -84,144 m
28, -89,833 m
29, -95,553 m
30, -101,297 m
31, -107,060 m
32, -112,836 m
33, -118,623 m
34, -124,417 m
35, -130,218 m
36, -136,023 m
37, -141,832 m
38, -147,643 m
39, -153,455 m
40, -159,270 m
41, -165,085 m
42, -170,901 m
43, -176,718 m
44, -182,535 m
45, -188,353 m
46, -194,171 m
47, -199,989 m
48, -205,807 m
49, -211,625 m
50, -217,444 m
51, -223,262 m
52, -229,081 m
53, -234,899 m
54, -240,718 m
55, -246,537 m

56, -252,355 m
57, -258,174 m
58, -263,992 m
59, -269,811 m
60, -275,630 m
61, -281,448 m
62, -287,267 m
63, -293,086 m
64, -298,904 m
65, -304,723 m
66, -310,542 m
67, -316,360 m
68, -322,179 m
69, -327,997 m
70, -333,816 m
71, -339,635 m
72, -345,453 m
73, -351,272 m
74, -357,091 m
75, -362,909 m

The dimensions X and Y are not used because the model is in one dimension.

The variables are the following.

gTr01, Dissolved inorganic carbon tendency. ($\text{mmol C m}^{-3} \text{ s}^{-1}$).

gTr02, Ammonium tendency. ($\text{mmol N m}^{-3} \text{ s}^{-1}$).

gTr03, Nitrite tendency. ($\text{mmol N m}^{-3} \text{ s}^{-1}$).

gTr04, Nitrate tendency. ($\text{mmol N m}^{-3} \text{ s}^{-1}$).

gTr05, Phosphate tendency. ($\text{mmol P m}^{-3} \text{ s}^{-1}$).

gTr06, Silicate tendency. ($\text{mmol Si m}^{-3} \text{ s}^{-1}$).

gTr07, Iron tendency. ($\text{mmol Fe m}^{-3} \text{ s}^{-1}$).

PAR, X.

PAR_ice, Photosynthetically active radiation (PAR) during the ice-covered period (January 1st to July 17th and November 24th to December 31st) at the centre of the cell of the vertical grid. 0 during the open water period. Use this value for the PAR during the ice-covered period. ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$).

PAR_ow, Photosynthetically active radiation (PAR) during the open water period (July 18th to November 23rd) at the centre of the cell of the vertical grid. PAR that would have been present if there was no ice during the ice-covered period. Use this value for the PAR during the open water period. ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$).

PARF, X.

PARF_ice, Photosynthetically active radiation (PAR) during the ice-covered period (January 1st to July 17th and November 24th to December 31st) at the top of the cell of the vertical grid. 0 during the open water period. ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$).

PARF_ow, Photosynthetically active radiation (PAR) during the open water period (July 18th to November 23rd) at the top of the cell of the vertical grid. PAR that would have been present if there was no ice during the ice-covered period. ($\mu\text{mol photons m}^{-2} \text{ s}^{-1}$).

TRAC01, Dissolved inorganic carbon concentration. (mmol C m^{-3}).

TRAC02, Ammonium concentration. (mmol N m^{-3}).

TRAC03, Nitrite concentration. (mmol N m^{-3}).

TRAC04, Nitrate concentration. (mmol N m^{-3}).

TRAC05, Phosphate concentration. (mmol P m^{-3}).

TRAC06, Silicate concentration. (mmol Si m^{-3}).

TRAC07, Total iron concentration. (mmol Fe m^{-3}).

TRAC08, Dissolved organic carbon. (mmol C m^{-3}).

TRAC09, Dissolved organic nitrogen. (mmol N m^{-3}).

TRAC10, Dissolved organic phosphorous. (mmol P m^{-3}).

TRAC11, Dissolved organic iron. (mmol Fe m^{-3}).

TRAC12, Particulate organic carbon. (mmol C m^{-3}).

TRAC13, Particulate organic nitrogen. (mmol N m^{-3}).

TRAC14, Particulate organic phosphorous. (mmol P m^{-3}).

TRAC15, Particulate organic silicon. (mmol Si m^{-3}).

TRAC16, Particulate organic iron. (mmol Fe m^{-3}).

TRAC17, Particulate inorganic carbon. (mmol C m^{-3}).

TRAC18, Alkalinity. (mmol eq m^{-3}).

TRAC19, Dissolved oxygen. (mmol O m^{-3}).

TRAC20, Chromophoric dissolved organic matter. (mmol P m^{-3}).

TRAC21, X.

TRAC22, X.

TRAC23, Picophytoplankton, Equivalent Spherical Diameter (ESD) 1.4 μm . (mmol C m^{-3}).

TRAC24, Picophytoplankton, ESD 2.0 μm . (mmol C m^{-3}).

TRAC25, Other nanophytoplankton, ESD 3.0 μm . (mmol C m^{-3}).

TRAC26, Other nanophytoplankton, ESD 4.5 μm . (mmol C m^{-3}).

TRAC27, Other nanophytoplankton, ESD 6.6 μm . (mmol C m^{-3}).

TRAC28, Other nanophytoplankton, ESD 10 μm . (mmol C m^{-3}).

TRAC29, Other nanophytoplankton, ESD 15 μm . (mmol C m^{-3}).

TRAC30, X.

TRAC31, X.

TRAC32, X.

TRAC33, X.

TRAC34, X.

TRAC35, Diatom, ESD 6.6 μm . (mmol C m^{-3}).

TRAC36, Diatom, ESD 10 μm . (mmol C m^{-3}).

TRAC37, Diatom, ESD 15 μm . (mmol C m^{-3}).

TRAC38, Diatom, ESD 22 μm . (mmol C m^{-3}).

TRAC39, Diatom, ESD 32 μm . (mmol C m^{-3}).

TRAC40, Diatom, ESD 47 μm . (mmol C m^{-3}).

TRAC41, Diatom, ESD 70 μm . (mmol C m^{-3}).
 TRAC42, Diatom, ESD 104 μm . (mmol C m^{-3}).
 TRAC43, Diatom, ESD 154 μm . (mmol C m^{-3}).
 TRAC44, Dinoflagellate, ESD 6.6 μm . (mmol C m^{-3}).
 TRAC45, Dinoflagellate, ESD 10 μm . (mmol C m^{-3}).
 TRAC46, Dinoflagellate, ESD 15 μm . (mmol C m^{-3}).
 TRAC47, Dinoflagellate, ESD 22 μm . (mmol C m^{-3}).
 TRAC48, Dinoflagellate, ESD 32 μm . (mmol C m^{-3}).
 TRAC49, Dinoflagellate, ESD 47 μm . (mmol C m^{-3}).
 TRAC50, Dinoflagellate, ESD 70 μm . (mmol C m^{-3}).
 TRAC51, Dinoflagellate, ESD 104 μm . (mmol C m^{-3}).
 TRAC52, Dinoflagellate, ESD 154 μm . (mmol C m^{-3}).
 TRAC53, Dinoflagellate, ESD 228 μm . (mmol C m^{-3}).
 TRAC54, Zooplankton, ESD 6.6 μm . (mmol C m^{-3}).
 TRAC55, Zooplankton, ESD 10 μm . (mmol C m^{-3}).
 TRAC56, Zooplankton, ESD 15 μm . (mmol C m^{-3}).
 TRAC57, Zooplankton, ESD 22 μm . (mmol C m^{-3}).
 TRAC58, Zooplankton, ESD 32 μm . (mmol C m^{-3}).
 TRAC59, Zooplankton, ESD 47 μm . (mmol C m^{-3}).
 TRAC60, Zooplankton, ESD 70 μm . (mmol C m^{-3}).
 TRAC61, Zooplankton, ESD 104 μm . (mmol C m^{-3}).
 TRAC62, Zooplankton, ESD 154 μm . (mmol C m^{-3}).
 TRAC63, Zooplankton, ESD 228 μm . (mmol C m^{-3}).
 TRAC64, Zooplankton, ESD 338 μm . (mmol C m^{-3}).
 TRAC65, Zooplankton, ESD 502 μm . (mmol C m^{-3}).
 TRAC66, Zooplankton, ESD 744 μm . (mmol C m^{-3}).
 TRAC67, Zooplankton, ESD 1103 μm . (mmol C m^{-3}).
 TRAC68, Zooplankton, ESD 1636 μm . (mmol C m^{-3}).
 TRAC69, Zooplankton, ESD 2425 μm . (mmol C m^{-3}).
 PAR and PARF are not used. Their values are always equal to 0. TRAC21, TRAC22, TRAC30, TRAC31, TRAC32, TRAC33 and TRAC34 correspond to phytoplankton that were removed from the model. Their values are always equal to 0.

- Files:
 - run_20240930_0000_EXP2_3si_yy_no3_xx_122_sims: simulation outputs for EXP-SyyNxx.
 - run_20240930_0000_EXP2_3si_standard_no3_standard_122_sims: simulation outputs for EXP-0.